Hongjun Jang

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Professional Summary

Staff Software Engineer with 6+ years of experience designing, implementing, and evaluating AI and knowledge systems for production and on-device environments. Hands-on experience in Agentic AI memory, Retrieval-Augmented Generation (RAG), natural-language-to-query, embeddings and vector search, knowledge graphs, SPARQL, LLM/SLM evaluation, and cross-platform systems. Technical Lead for cross-organizational research initiatives in on-device RAG, automated knowledge graph generation, and NL-to-Query, with experience translating technical requirements into architectures and evaluation criteria across accuracy, latency, memory, storage, token efficiency, and deployability.

Experience

- **Samsung Research, HQ** — Seoul, South Korea
  Staff Software Engineer — Mar. 2026 – Present
  - **Agentic AI Memory and Retrieval**: Designed and developed the graph retrieval layer of an on-device agentic memory platform, covering graph-based retrieval, semantic and vector search, full-text retrieval, and LLM-based entity and fact extraction.
  - **Technical Lead, On-Device RAG and Knowledge Generation**: Lead cross-organizational research with Samsung AI Center Warsaw on SLM-based on-device RAG and automated knowledge graph generation, defining technical direction, system architecture, evaluation methodology, and development priorities.
  - **Technical Lead, On-Device NL-to-Query**: Lead research with Samsung Research UK on on-device natural-language-to-query systems for SQL and SPARQL, focusing on accurate and efficient structured data access under resource-constrained deployment environments.
- **Samsung R&D Institute UK** — Oxford, United Kingdom
  Software Engineer — Oct. 2024 – Feb. 2026
  - **Secure Knowledge and Retrieval Architecture**: Designed and implemented OAuth 2.0 authentication for federated SPARQL queries in a commercial graph database engine, including concurrency-safe token management, expiry handling, and secure remote data access.
  - **Production and Cross-Platform Engineering**: Built Azure-based CI/CD pipelines and stable C/C++ APIs using the Hourglass pattern to support reliable deployment of a commercial graph database engine across heterogeneous platforms.
  - **Cross-Organization Collaboration**: Collaborated with Samsung Research engineering teams and Oxford Semantic Technologies to translate product requirements into extensible graph database, federated query, and reasoning capabilities for production environments.
- **Samsung Research, HQ** — Seoul, South Korea
  Software Engineer — Jul. 2020 – Sep. 2024
  - **On-Device Knowledge Platform**: Developed a core module of the Personal Data Engine for Samsung Galaxy devices, integrating an embedded knowledge graph database and inter-application communication to enable on-device reasoning and personalized data services.
  - **AI Optimization with Reinforcement Learning**: Designed and implemented multi-objective reinforcement learning algorithms for decision optimization under conflicting operational objectives; demonstrated approximately 20% power savings in a base-station PoC conducted with a leading U.S. telecom operator.

Technical Skills

- **AI / ML**: Agentic AI, Retrieval-Augmented Generation (RAG), LLM/SLM Evaluation, Embeddings, Vector Search, Semantic Search, Prompt and Context Engineering, Reinforcement Learning
- **Knowledge and Retrieval Systems**: Knowledge Graphs, RDF, SPARQL, NL-to-SQL, NL-to-SPARQL, Graph Reasoning, RDFox, Full-Text Search, Information Retrieval
- **Programming**: C++20, Python, Rust, Kotlin, Java
- **Systems / Cloud**: Linux, Android, Azure Pipelines, Ansible, Ray, SQLite, Google Cloud BigQuery

Education

- **Korea University** — Seoul, South Korea
  Bachelor of Science in Computer Science — Mar. 2013 – Jul. 2020
  GPA: 3.88 / 4.5 (93.8 / 100)

Patents and Publications

- **Hongjun Jang**, Eunkyu Oh, Minsoo Kim, Taehun Kim, Yunhu Ji, "Method and Electronic Device for Providing Information Using Reinforcement Learning," US Patent US20240221018A1.
- Minsuk Choi, Kyungrae Kim, **Hongjun Jang**, Hyokyung Woo, Seowoo Jang, "Electronic Device and Operation Method for Deciding Cell On/Off Threshold Based on Traffic," US Patent US20230032030A1.
- Minsuk Choi, Kyungrae Kim, **Hongjun Jang**, Hyokyung Woo, Joan S. Pujol-Roig, Yue Wang, Hunje Yeon, Sunghyun Choi, Seowoo Jang, "Cell On/Off Parameter Optimization for Saving Energy via Reinforcement Learning," IEEE Globecom Workshops (GC Wkshps), 2021.

Selected Awards and Honors

- **Samsung AI Top 30 (AI Young Leader), 2026**: Selected as one of 30 company-wide AI Young Leaders.
- **Samsung Research S/W Award for Individual, 2024**: Individual technical achievement award.
- **Samsung S/W Certificate: Expert Programmer, 2020**
- **International Collegiate Programming Contest, Ho Chi Minh City Regional, 2017**: 3rd Prize
- **International Collegiate Programming Contest, Daejeon Regional, 2016**: 4th Place; Minister of Science, ICT and Future Planning Award

Technical Leadership and Teaching

- **International Olympiad in Informatics (IOI)**: Student Assistant Coach for the Korean national team (2016); Coach, IOI Training Camp in Korea (2013–2017).
- **Korea Olympiad in Informatics (KOI)**: Coach and Problem Setter (2013–2017).